



Review

A systematic survey of air quality prediction based on deep learning

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ABSTRACT

The impact of air pollution on public health is substantial, and accurate long-term predictions of air quality are crucial for early warning systems to address this issue. Air quality prediction has drawn significant attention, bridging environmental science, statistics, and computer science. This paper presents a comprehensive review of the current research status and advances in air quality prediction methods. Deep learning, a novel machine learning approach, has demonstrated remarkable proficiency in identifying complex, nonlinear patterns in air quality data, yet its application in air quality prediction is still relatively nascent. This paper also conducts a systematic analysis and summarizes how cutting-edge deep learning models are applied in air quality prediction. Initially, the historical evolution of air quality prediction methods and datasets is presented. This is followed by an examination of conventional air quality prediction techniques. A thorough comparative analysis of progress made with both traditional and deep learning-based prediction methods is provided. This review particularly focuses on three aspects: temporal modeling, spatiotemporal modeling, and attention mechanisms. Finally, emerging trends in the field of air quality prediction are identified and discussed.

1. Introduction

Undoubtedly, air quality significantly impacts individuals' well-being and daily lives. The continuous advancement of social science and technology, coupled with the relentless progression of commercialization, inevitably leads to fluctuations in air quality. Consequently, this matter has drawn extensive attention from both the public and relevant governmental bodies. In response, numerous countries have recently initiated the establishment of extensive networks of air quality monitoring stations. On October 17, 2013, the International Agency for Research on Cancer (IARC), a division of the World Health Organization (WHO), officially classified outdoor air pollution as carcinogenic to humans [1]. Consequently, there is growing research interest in air quality prediction within the fields of environmental science, statistics, and computer science. Currently, reducing carbon emissions, improving air quality, and addressing climate change have become primary challenges hindering economic and social development. Accurate air quality prediction is a crucial component of meteorological services, and the continuous management of air quality plays a pivotal role in assessing pollutant emission trajectories and changes in air quality. Enhancing the

accuracy of air quality prediction has the potential to accelerate the achievement of clean air and carbon neutrality objectives. Therefore, there is an urgent need to fully harness the supporting role of machine learning models in air quality prediction tasks.

Accurately predicting air quality presents significant challenges. Air quality forecasting relies on diverse data sources, including ground monitoring stations, meteorological satellites, and monitoring aircraft. However, these data sources contain pollutant samples that exhibit high spatial and temporal heterogeneity. This heterogeneity poses challenges in aggregating relevant data to achieve accurate predictions of pollutant changes. Among these data sources, ground air quality monitoring stations play a vital role because they are strategically positioned near areas with high pollutant emissions. This proximity allows them to serve as crucial data sources for studying air quality variations. Nevertheless, several challenges persist in the field of air quality prediction. First, despite some regions successfully establishing numerous regional air quality monitoring stations and even deploying interconnected micro air quality monitoring stations on streets, many areas still lack a comprehensive air quality monitoring system. This approach results in limited and sparse data samples in certain regions. Second, the generation of air

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pollutants involves complex chemical processes that exhibit significant temporal and spatial variations. In this context, predicting air quality solely through a fixed formula becomes challenging. Therefore, achieving accurate air quality prediction remains a formidable issue due to the complexity of air pollution dynamics and constraints in terms of data availability and aggregation.

In recent years, the aforementioned issues have given rise to two categories of commonly used methods for air quality prediction: traditional prediction methods based on hand-designed features and machine learning algorithms based on deep learning. Traditional prediction methods include numerical modeling methods, statistical approaches, and traditional machine learning techniques [2]. Numerical modeling methods are built upon atmospheric dynamics and chemical processes, entailing the construction of mathematical models to simulate pollution data, either horizontally or vertically, by incorporating meteorological data. However, statistical methods place less emphasis on the physico-chemical properties of air pollutants and make air quality predictions by employing statistical models that directly integrate meteorological data with prior air quality data. While these methods effectively integrate multi-domain knowledge, they also possess limitations. For instance, flexible multiscale frameworks are lacking when handling large datasets, and extensive computational operations may lead to errors. Furthermore, due to their linearity constraints, they struggle to address nonlinear problems. To overcome these challenges, researchers have turned to the utilization of nonlinear machine learning approaches, such

as support vector regression (SVR) and random forest regression (RFR) [3–5]. Despite the favorable performance of these nonlinear machine learning methods in air quality prediction, many of them face difficulties in capturing complex time series patterns of air quality data and modeling the long-term impact of air pollution. Consequently, this limitation can negatively affect the prediction accuracy.

In recent years, deep learning techniques have emerged, attracting significant amounts of attention across various fields [6,7]. Deep learning methods effectively capture underlying patterns within samples and extract high-level features, demonstrating remarkable success in various domains such as computer vision, speech signal processing, and natural language processing [8–10]. However, there is a paucity of systematic surveys on recent deep learning-based air quality prediction methods. To rectify this gap, this paper offers a comprehensive review of the current research status and advancements in air quality prediction methods. Fig. 1 illustrates the evolution of air quality prediction methods and associated datasets. This review commences by examining domestic and international air pollution datasets, followed by an introduction to conventional air quality prediction methods. Subsequently, air quality prediction approaches grounded in deep learning techniques, including representative deep learning methods, temporal and spatiotemporal modeling, and attention mechanisms, are systematically summarized. Furthermore, this review provides a detailed analysis of research progress spanning from 2010 to 2023. Finally, this paper underscores the challenges within this domain and outlines future

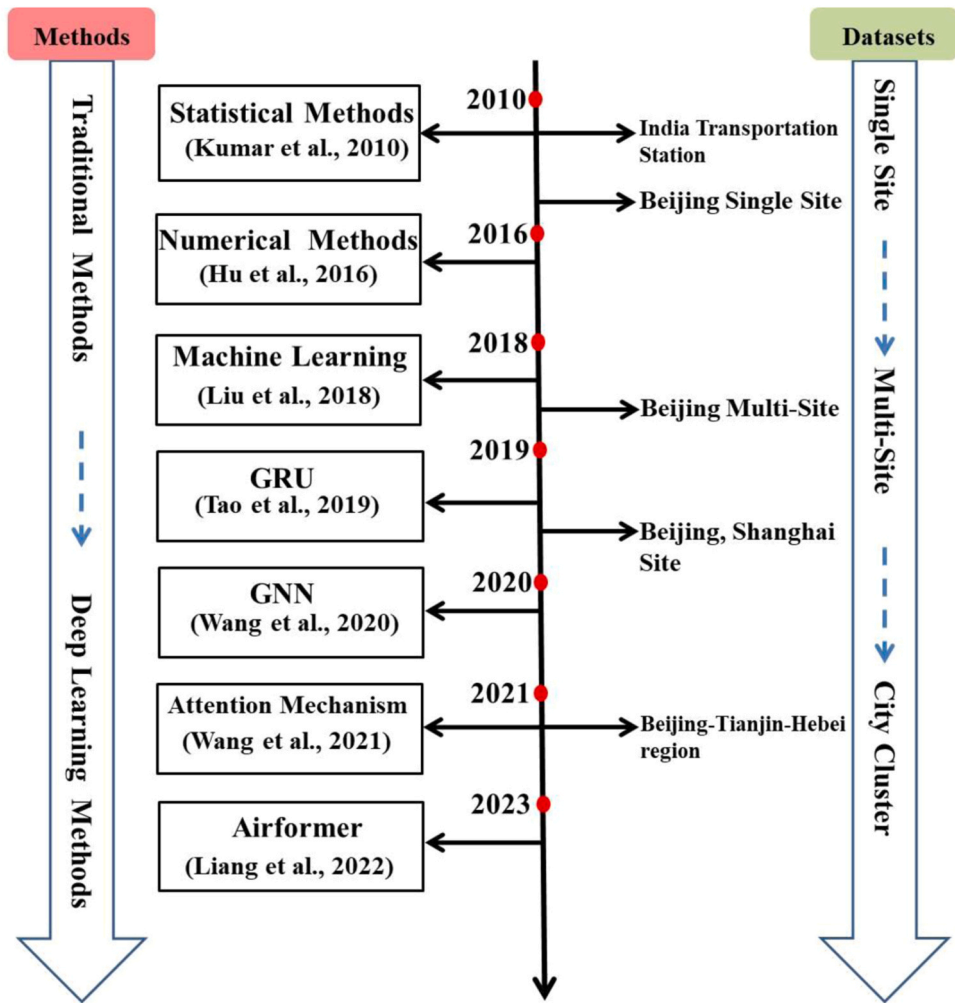


Fig. 1. The evolution process of air quality prediction methods and relevant datasets. From 2010–2023, the development of air quality prediction algorithms has undergone two distinct stages, shifting from traditional methods to the utilization of deep learning techniques. Simultaneously, air quality datasets have evolved from being limited to single air quality monitoring sites to encompassing multiple sites and city clusters.

directions for development.

For this review, a comprehensive and systematic literature search included popular academic search engines such as Google Scholar and Web of Science, along with reputable publishers such as IEEE, Springer, Elsevier, Wiley, ACM, Taylor & Francis, *etc.* The search included the time frame from 2010 to 2023, ensuring inclusion of the most current research literatures. A combination of keywords, including "air quality," "deep learning," "attention mechanisms," "temporal," and "spatiotemporal", was used to facilitate a comprehensive search. This culminated in the collection of more than 150 relevant literature sources from international journals and conferences in the fields of environment and computer science.

The primary contributions of this work are summarized as follows:

- (1) Summarizing the evolution of air quality prediction from traditional methods to deep learning methods by analyzing key milestones.
- (2) Providing an in-depth analysis of existing deep learning-based air quality prediction methods, including relevant datasets, typical deep learning techniques, and their applications in air quality prediction.
- (3) Discussing future research directions for deep learning-based air quality prediction.

The subsequent sections of this paper are structured as follows: [Section 2](#) offers an overview of the air quality datasets. [Section 3](#) describes air quality prediction utilizing traditional methods. [Section 4](#) reviews deep learning theories and typical deep learning methods. [Section 5](#) addresses research challenges and open issues within this field. Finally, [Section 6](#) presents the conclusions.

2. Air quality datasets and evaluation metrics

2.1. Datasets

To perform air quality prediction tasks, pertinent air quality datasets are indispensable. Researchers have compiled a variety of air quality datasets, including different types, for this purpose, as outlined in [Table 1](#).

A cursory review of [Table 1](#) reveals that the majority of these datasets originate from regional air quality monitoring stations. Typically, these datasets include measurements of various air pollutants, comprising fine particulate matter (PM_{2.5}), carbon monoxide (CO), sulfur dioxide (SO₂), nitrogen dioxide (NO₂), and ozone (O₃), in addition to meteorological data such as atmospheric temperature, atmospheric pressure, relative humidity, precipitation, and other pertinent parameters. The temporal granularity of these datasets commonly operates at an hourly level.

2.2. Evaluation metrics

To evaluate the air quality prediction performance, several typical metrics, including the mean absolute error (MAE), root mean square error (RMSE), and mean absolute percentage error (MAPE), are usually employed, as described below.

$$\text{MAE}(\hat{y}, y) = \frac{1}{m} \sum_{i=1}^m |y_i - \hat{y}_i| \quad (1)$$

$$\text{RMSE}(\hat{y}, y) = \sqrt{\frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2} \quad (2)$$

$$\text{MAPE}(\hat{y}, y) = \frac{1}{m} \sum_{i=1}^m \frac{|y_i - \hat{y}_i|}{y_i} \quad (3)$$

where y and $\hat{y}$ represent the ground-truth and predicted PM_{2.5} values,

Table 1

Representative air quality datasets.

Location	Period	Data Sources	Brief Introduction
Beijing Single Site[2]	01/01/2010–12/31/2014	U.S. Embassy in Beijing	Hourly PM _{2.5} and meteorological data from a single site in Beijing, China
Beijing Multi-Site[11]	01/01/2015–12/31/2016	China National Environmental Monitoring Center	Hourly PM _{2.5} , PM ₁₀ and other air quality data from 12 stations in Beijing, China
Beijing Multi-Site[12]	01/01/2017–05/01/2018	Beijing Air Quality Monitoring Station	Hourly PM _{2.5} , PM ₁₀ and other air quality data from 35 stations in Beijing, China
Shanghai Multi-Site[13]	03/01/2015–02/28/2017	Shanghai Environmental Monitoring Center	Hourly PM _{2.5} , PM ₁₀ and other air quality data from 2 stations in Shanghai, China
Huaihai Economic Zone[14]	03/01/2017–02/28/2019	China National Ministry of Ecology and Environment	Hourly PM _{2.5} , PM ₁₀ and other air quality data from 10 cities in the Huaihai Economic Zone, China
Lanzhou Multi-Site[15]	11/01/2017–01/01/2019	Lanzhou air micro-monitoring station	Hourly PM _{2.5} , PM ₁₀ and other air quality data from 359 stations in Lanzhou, China
Shijiazhuang Single Site[16]	01/01/2019–12/31/2019	Shijiazhuang Air Monitoring Station	Hourly PM _{2.5} , PM ₁₀ and other air quality data from a single site in Shijiazhuang, China
U.S. States[17]	01/01/2000–12/31/2016	U.S. Environmental Protection Agency	Hourly PM _{2.5} , PM ₁₀ and other air quality data in US
Derry Multi-Site[18]	12/01/2018–11/30/2019	Central Pollution Control Board of India	Hourly PM _{2.5} , PM ₁₀ , and meteorological data from Delhi, India

respectively, and m denotes the total number of test data. The consensus in the field generally acknowledges that lower values of MAE and RMSE indicate superior prediction results. These metrics evaluate the average magnitude of errors and furnish a comprehensive gauge of the model's predictive precision. A diminished MAE or RMSE implies that the model's predictions closely align with the actual values, underscoring heightened accuracy. In contrast, the MAPE serves as a relative error metric, portraying prediction errors as a percentage of the actual value. The MAPE exhibits particular sensitivity to outlier data and tends to yield elevated values when extreme or diminutive actual values are encountered.

3. Air quality prediction with traditional methods

Early methods for air quality prediction primarily include numerical modeling, statistical methods, and traditional machine learning approaches. Numerical modeling methods resemble traditional weather forecasting methods, in which temporal and spatial dimensions are partitioned into grids based on atmospheric physics and chemistry. These methods employ computer simulations to predict meteorological data and air pollution levels. Common numerical modeling methods include the Community Multiscale Air Quality Modeling System (CMAQ), Comprehensive Air Quality Model with Extensions (CAQME), and Nested Air Quality Prediction Modeling System (NAQPMS) [19–23]. Statistical methods often rely on assumptions such as linear dependence and stationarity, constructing mathematical models through curve fitting and parameter estimation. Common statistical models include the autoregressive moving average (ARMA), autoregressive integrated moving average (ARIMA), multivariable linear regression (MLR), and time series regression [24–28]. Classical machine learning methods mitigate these assumed errors by configuring specific parameters for air quality prediction tasks. Widely-used classical machine learning

methods include SVR and RFR.

Hu et al. conducted a year-long simulation of air quality prediction in China using Weather Research and Forecasting (WRF) and CMAQ models [29,30]. The experimental results demonstrate the effectiveness of CMAQ models in reproducing PM_{2.5} and O₃ concentrations in most cities, accurately capturing air pollution patterns in China, and offering insights for potential improvements. Kumar et al. employed ARMA, ARIMA, and various other models to forecast air pollution data from a traffic station in Delhi, India [31]. This approach determines the autoregressive and moving average (MA) parameters of the models through different information criteria, such as the Bayesian information criterion (BIC) [32]. Cheng et al. proposed an MA-OWA model based on the MA and order weighted average (OWA) [33–35]. The MA-OWA model effectively calculates ozone concentrations at urban air stations in Taiwan through a weighted aggregation of the OWA operator and simplified MA calculation, outperforming traditional models such as MA and ARMA. Lei et al. proposed the application of Classification and Regression Trees (CART) and Multiple Regression (MR) to establish a multipollutant statistical model [36–38]. The prediction results displayed a high level of confidence in forecasting air quality conditions in Macau. Liu et al. compared the results of conventional machine learning approaches, including MLR, RFR, and SVR, for predicting the air quality index (AQI) [39]. The experiments indicated that RFR, due to the interaction of variables within the random forest model, exhibited superior generalizability and greater accuracy in predicting AQIs, particularly in the Beijing region. Nieto et al. utilized SVR to develop a robust nonlinear dynamic model for air quality prediction, specifically in Spanish cities [40]. The presented model effectively captured the interdependence of major pollutants within the study sites, demonstrating its remarkable ability for nonlinear modeling. Kumar et al. applied ARIMA, principal component regression (PCR), and a comprehensive combined model to forecast the daily AQI for each season in Delhi, India [41–43]. The experimental results highlighted that integrating ARIMA with PCR effectively handled the autocorrelation of air variables. Furthermore, the incorporation of meteorological parameters based on principal component analysis (PCA) significantly enhances the accuracy and stability of air quality prediction tasks [44,45].

3.1. Highlights in this section

The efficacy of conventional air quality prediction methods is heavily contingent on the precision of the air pollution data and the comprehensiveness of the employed models. Nevertheless, these methods are encumbered by specific limitations. Both of these methods necessitate extensive computations and costly hardware or rely on a substantial historical air quality dataset for training to improve prediction accuracy. Consequently, there has been a recent shift toward exploring air quality prediction models characterized by heightened adaptability concerning both data and hardware requisites.

4. Deep learning theories

Deep learning [46], a subset of machine learning, provides a potent and well-suited approach for managing large datasets. This approach emulates the complex processes of the human brain to handle diverse data modalities, including images, audio, and text. Its remarkable accomplishments span various domains, including computer vision, natural language processing, and speech signal processing [47–49]. Deep learning excels in extracting high-level features from extensive data. Specifically, air quality prediction has emerged as a prominent research area where deep learning has proven its efficacy in achieving robust prediction performance. Representative deep learning models include the deep belief network (DBN), convolutional neural network (CNN), recurrent neural network (RNN), and so on [50–52], as described below.

4.1. Deep belief network

A Deep Belief Network (DBN) [50] represents a type of deep neural network that distinguishes itself as a probabilistic generative model, unlike traditional discriminative neural networks. The classic structure of a DBN comprises multiple layers of restricted Boltzmann machines (RBMs) and a back propagation (BP) layer. An RBM, a variant of the neural perceptron, consists of two layers: the visible layer containing explicit elements and the hidden layer comprising hidden elements. The visible layer serves as the input for training the data, while the hidden layer operates as a feature detector. The neurons within these two layers are complexly interconnected in a bidirectional, fully connected (FC) manner.

The training process for the DBN model typically includes two distinct stages: pretraining and fine-tuning. In the pretraining stage, each layer within the RBM network undergoes independent and unsupervised training. The primary objective of this stage is to identify an optimal weight configuration that maximizes the probability of generating training samples. In the subsequent fine-tuning stage, the BP network employs the feature vector generated by the RBM as input. While training the classifier, error propagation occurs from the topmost to the lowest layer of the RBM, facilitating comprehensive refinement of the entire DBN network. Due to the robust flexibility of DBNs and their compatibility with parallel computing, their applications have gained prominence, particularly in domains such as image processing and speech signal processing [53,54].

Lee et al. proposed a convolutional deep belief network (CDBN), which enables the DBN model to extract high-dimensional features through translation invariance based on maximum probability pooling [55]. This method addresses the limitation of the DBN network's inability to extend its formulation to real-world visible units. Zhao et al. presented a WTADBN network that leveraged the winner-take-all (WTA) concept for stratifying lung cancer patients [56,57]. This approach effectively resolves the issue of feature homogeneity inherent in the DBN network.

4.2. Convolutional neural network

A Convolutional Neural Network (CNN) [52] stands out as one of the most successful and widely adopted deep learning methodologies because of its robust data processing capabilities. The CNN structure usually contains 1D, 2D, and 3D variants. Among these, the 1D CNN proves invaluable for processing time series and audio signals. In contrast, 2D CNNs excel in tasks related to text recognition and image analysis, while 3D CNNs aptly capture spatiotemporal features, demonstrating extensive use in video analysis.

The fundamental CNN structure comprises five key layers: the input layer, convolutional layer, pooling layer, FC layer, and output layer. The input layer handles multidimensional data, including 1D, 2D, or 3D data. The convolutional layer consists of multiple convolutional units, progressively extracting features from low to high levels within the input data through iterative convolution operations. The parameters of the convolutional layers include the size of the convolution kernel, the convolution step, and the padding method. Subsequently, the pooling layer operates on the feature maps obtained from the convolutional layer's feature extraction. It subdivides the post-convolutional features into regions, reducing dimensionality through pooling operations. The FC layer nonlinearly combines the extracted features to obtain the output. The output layer aligns with other neural networks, using the softmax function for image classification tasks to produce label outputs or directly generate numerical results for numerical regression.

At present, numerous researchers [58] continue to enhance and diversify CNN architectures. Representative CNN models include AlexNet [59], ResNet [60], GoogleNet [61], and VGG [62]. Notably, ResNet [60] is the most widely adopted CNN model. Its distinctive residual network structure, featuring a "short circuit" connection between input

and output, effectively addresses challenges related to network degradation and gradient vanishing.

It is worth noting that conventional CNNs often possess a large number of network parameters, resulting in high computational complexity. To mitigate this issue, several studies have concentrated on developing lightweight CNNs. Zhang et al. proposed ShuffleNet [63], a computationally efficient lightweight CNN model that utilizes point-by-point grouped convolution and channel shuffling to significantly reduce computational costs while maintaining performance. Wang et al. presented a compact convolution module called DSC-Ghost-Conv [64], which combines the advantages of two convolutional techniques: depthwise separable convolution and the ghost convolution module. This module can be seamlessly integrated into traditional CNN architectures such as ResNet and VGG.

4.3. Recurrent neural network

Recurrent Neural Network (RNN) [51] operate cyclically, enabling them to incorporate information from both the current and previous moments in the input sequence. This unique property makes RNNs a preferred choice for tasks involving sequential data, such as sequence data modeling and natural language processing. However, when confronted with long-term sequence data, RNNs may struggle to effectively capture temporal dependencies, potentially leading to issues such as gradient explosion or vanishing gradients.

To address this challenge, a variant of RNN known as long short-term memory (LSTM) [65] has emerged. LSTM, as a specialized form of RNN, excels at learning long-term dependencies in sequential data. The core component of LSTM consists of three distinct nonlinear gating mechanisms: the input gate, forget gate, and output gate. These gates operate selectively, allowing for the retention or omission of specific input information. This controlled modulation significantly enhances the ability of LSTM to process long-term sequence data. The gated recurrent neural network (GRU) [66], on the other hand, represents a simplified version of LSTM, where the input gate and forget gate are combined into a single update gate. Compared to LSTM, the GRU stands out for its efficiency in both training and computational requirements, which are achieved through parameter reduction. It is worth noting that the GRU tends to outperform the LSTM in scenarios with limited data. However, in situations with ample available data, the effectiveness of LSTM may remain superior. Further details on the various LSTM variants can be found in [67].

5. Air quality prediction based on deep learning

The modeling of air pollutant samples represents a pivotal aspect of air quality prediction research. Currently, two primary categories of deep learning models predominate in the domain of air quality prediction. The first category includes temporal modeling, leveraging RNNs, long short-term memory (LSTM), and other temporal deep networks. The second category is geared toward capturing spatiotemporal information from sequence data and is achieved through the integration of CNNs with RNNs/LSTMs and other spatiotemporal network architectures. Additionally, recent deep learning methodologies, enhanced by attention mechanisms such as the Transformer [68], have made their way into this field and have attracted considerable interest in recent years.

5.1. Air quality prediction with temporal modeling

When considering temporal information for air quality prediction, it is common practice to treat air pollution data as one-dimensional time series or signals. To prepare the data for subsequent analysis, various preprocessing techniques, including the Discrete Fourier Transform (DFT) [52] and wavelet transform [69], are applied. Following preprocessing, the data are fed into deep learning models to extract

temporal features for the final air quality prediction. Prominent deep learning models in this context include RNNs, LSTMs, GRUs, and 1D CNNs.

Zhang et al. presented a hybrid deep learning approach named VMD-BiLSTM [70], as illustrated in Fig. 2. This model combines variational mode decomposition (VMD) [71] with bidirectional long short-term memory (BiLSTM) to predict PM_{2.5} levels in Beijing. VMD is employed to decompose the initial PM_{2.5} data into subsignal components, reducing volatility and nonlinearity while smoothing the data. A BiLSTM is subsequently used to predict each component separately, resulting in improved prediction accuracy. The experimental results highlight the effectiveness of BiLSTM in capturing temporal dependencies within PM_{2.5} sequence data. Jin et al. proposed an innovative deep learning framework called MTMC-NLSTM that incorporates multiple nested long short-term memory networks (NLSTMs) [72], as shown in Fig. 3. This framework divides initial data into denoised low-frequency and high-frequency components using the wavelet transform. These processed data are input into the NLSTM to predict AQI values across 12 sites in Beijing [72,73]. The results demonstrate significant advantages in predicting multivariate air quality trends over time, with the added benefit of short training times, enabling near real-time AQI tracking. Luo et al. developed an ARIMA-WOA-LSTM model for air pollutant prediction tasks [74]. This model utilizes ARIMA to extract linear components of air pollution data and applies a WOA-LSTM model to forecast nonlinear components based on the whale optimization algorithm (WOA), optimizing hyperparameters for LSTM.

Tao et al. presented the CBGRU model for PM_{2.5} prediction, which combines the capabilities of a 1D CNN and a bidirectional gated recurrent unit network (BiGRU) [75]. This model simplifies data with a CNN for downsampling and captures temporal features with BiGRU for prediction. Retta et al. conducted a comparative analysis of various networks, including LSTM, GRU, and ConvLSTM [76,77]. The results indicate that temporal convolutional networks (TCNs) perform comparably to long short-term memory (LSTM) and gated recurrent units (GRUs) networks when handling time series sequence data. Pruthi et al. integrated neural networks, fuzzy inference, and wavelet transforms for forecasting PM_{2.5} concentrations [78], achieving accurate predictions at a 1-kilometer resolution in less computational time. Fang et al. presented an air quality prediction approach based on an improved vanilla long short-term memory (LSTM) model that incorporates multichannel input and multiroute output (IVLSTM-MCMR) [79]. The experimental results on real datasets demonstrated the model's strong prediction performance.

5.2. Highlights in this section

Air quality prediction relying solely on temporal information can achieve satisfactory performance. Nevertheless, air pollution data exhibit notable temporal characteristics and substantial spatial interdependence, largely due to the collection of data from closely situated air quality monitoring stations. Neglecting spatial information could lead to decreased accuracy and reliability in air quality predictions. Consequently, the integration of spatiotemporal information is essential for enhancing the performance of air quality predictions.

5.3. Air quality prediction with spatiotemporal modeling

Given the spatiotemporal attributes inherent in air pollution data, the incorporation of spatial information into deep learning models has consistently led to improved prediction accuracy [81]. Consequently, a significant portion of contemporary deep learning approaches for air quality prediction simultaneously addresses both spatial and temporal dependencies embedded within time series sequence data. The typical deep neural networks predominantly involved in spatial modeling include DNNs, CNNs, and graph neural networks (GNNs) [82]. Moreover, the introduction of convolutional operations in GNNs, known as

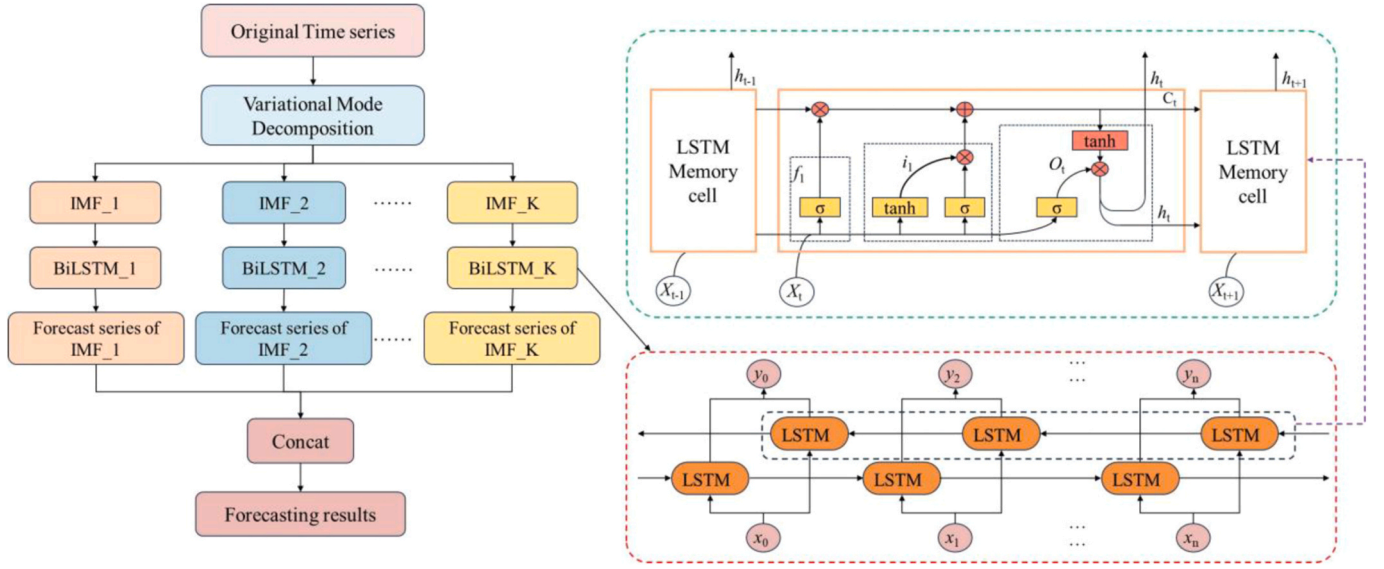


Fig. 2. The flow chart of VMD-BiLSTM[70].

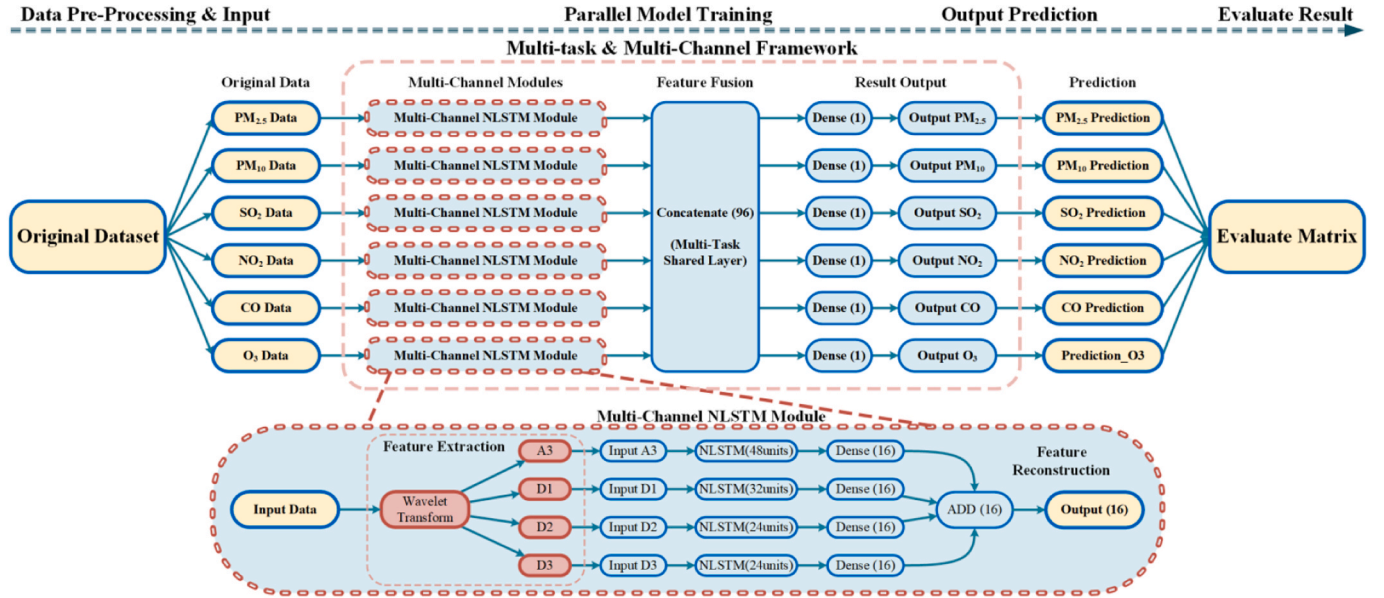


Fig. 3. The structure of the proposed MTMC-NLSTM model[72].

graph convolutional neural networks (GCNs), has gained significant attention due to its ability to effectively capture spatial information in recent years [83,84]. Models that integrate both spatial and temporal modeling include CNN-LSTM, CNN-GRU, and GCN-LSTM [85].

Han et al. introduced a deep model called the multi-adversarial spatiotemporal recurrent graph neural network (MasterGNN) for predicting air quality and weather conditions [80]. The architecture of this model, as illustrated in Fig. 4, incorporates a recurrent GNN to capture the spatiotemporal relationships between air quality and monitoring stations. An adversarial graph learning strategy is employed to counteract noise arising from spatiotemporal modeling, and multi-adversarial learning is adopted to expedite the network's training process. The experimental results highlight the significant enhancement in prediction accuracy achieved through the integration of graph neural networks and adversarial learning. Wu et al. developed a multiscale spatial-temporal network (MSSTN) for air quality prediction [86]. This network comprises three subnetworks: a dilated convolutional network

for extracting features across various time scales, a designed GCN for capturing spatial characteristics, and a fusion network for combining these extracted features to generate accurate prediction results. The cooperative efforts of these three networks enable the MSSTN model to adequately capture complex multiscale spatiotemporal patterns and interregional interactions, leading to improved prediction performance. Huang et al. proposed a predictive model named SpAttRNN, which integrates spatial attention embedding with a recurrent neural network [87]. This model captures the temporal and spatial characteristics of each air monitoring station separately, allowing for the modeling of dynamic spatiotemporal correlations between the stations. To account for spatial correlation, the SpAttRNN model incorporates a normalized adjacency matrix that includes geographic distance and geometric relationships between the target and neighboring stations. Additionally, an attention matrix is utilized to emphasize significant correlations among observation stations. The SpAttRNN model further employs LSTM to encode original temporal features from target sites and

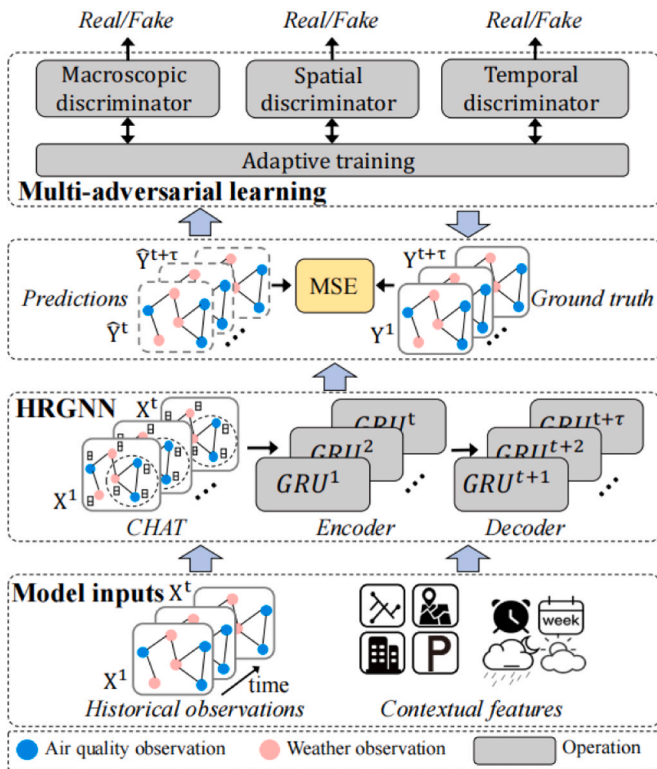


Fig. 4. The flowchart of MasterGNN[80].

generates final prediction results by combining the attention matrix with LSTM. Yeo et al. employed a five-layer CNN architecture along with a three-layer GRU model to generate real-time predictions for $PM_{2.5}$ across 25 monitoring stations in the Seoul region [88]. They also optimized computational efficiency by using a multilayer model and proposed a geographic model to determine the optimal count of necessary neighboring sites, thereby enhancing prediction accuracy. Li et al. developed a geointelligent model named Geoi-DBN for air pollution prediction [89]. This model integrates geographic correlation information with a DBN, and the results demonstrate its effectiveness for large-scale air pollution detection. Xiao et al. developed a dual-path dynamic directed graph convolutional network (DP-DDGCN) for $PM_{2.5}$ prediction [90]. This method employs a dynamic directed graph to learn spatial correlations dynamically and utilizes GRUs to extract inherent temporal correlations from time series sequence data. Gilik et al. presented a hybrid deep learning framework that predicts multiple pollutants by combining a CNN and LSTM [91]. Faraji et al. developed a spatiotemporal prediction model tailored for $PM_{2.5}$ in Tehran [92], which combines three-dimensional CNNs with GRUs, resulting in good prediction performance across multiple sites in Tehran. Huang et al. employed a hybrid air quality prediction model that integrates improved particle swarm optimization (IPSO), empirical mode decomposition (EMD), and LSTM [93]. The experimental results on the Beijing dataset indicate that this model achieves higher prediction accuracy. Zhao et al. explored AQI prediction while considering the effects of China's lockdown policy during the COVID-19 pandemic [94]. Utilizing a deep learning framework, they synergized temporal autocorrelation with spatial correlation, leading to good predictive achievements. Ge et al. proposed a multiscale spatiotemporal graph convolution network (MST-GCN) [95], which contains both a graph convolutional layer and a temporal convolutional layer to model spatial and long-term dependencies. Wu et al. proposed a hybrid deep learning framework that integrated ResNet, GCN, and BiLSTM [96] for short-term forecasting of NO_2 and O_3 . Lin et al. predicted short-term $PM_{2.5}$ concentrations through a geo-context-based diffusion convolutional recurrent neural

network (GC-DCRNN) [97]. This method captures spatial relationships by employing a geographic environmental similarity graph and utilizes its impact on air quality prediction. The experimental results verify the validity of the model. Wang et al. presented a long-term $PM_{2.5}$ prediction model called the $PM_{2.5}$ -GNN model, which incorporates domain knowledge of $PM_{2.5}$ and employs a combination of GNNs and GRUs [98]. The model uses a GNN to capture the lateral transport of pollutants and integrates the GRU model to simulate the vertical accumulation process of pollutants. The experimental results show that this model achieves satisfactory prediction performance across extended time scales. Elbaz et al. combined a ResNet model with a convolutional long short-term memory (ConvLSTM) model for air quality forecasting tasks using data from the city of NEOM, Saudi Arabia [99]. The ConvLSTM approach was employed to improve the model performance by using a ResNet model for learning high-level spatial features from meteorological and pollutant data.

5.4. Highlights in this section

Incorporating spatiotemporal information into air quality prediction allows for the identification of local variation trends and the discovery of long-term spatiotemporal dependence patterns in multivariate air pollution datasets. This approach has emerged as one of the predominant methods for air quality prediction. However, despite the widespread adoption of spatiotemporal deep learning models, these models have yet to fully leverage the potential of combination with traditional neural networks. Consequently, there is a pressing need to explore the applications of novel deep learning models in the context of air quality prediction to further advance the field.

5.5. Air quality prediction based on attention mechanisms

In recent years, recurrent network models such as RNNs and LSTM have demonstrated notable performance in various tasks. Nevertheless, owing to their sequential nature, they face challenges when parallelizing training samples. As a result, processing long-term sequences in batches frequently gives rise to memory constraints. Inspired by human visual attention mechanisms, attention models have overcome these limitations [101]. In contrast to recurrent network models, attention models can accommodate inputs of varying shapes and mitigate the challenge of imbalanced computing resource allocation. Consequently, attention models have gained substantial popularity and become one of the favored deep learning models. Among these attention models, the Transformer [68] is a novel network architecture based on the self-attention mechanism [93] and has been successfully applied to various tasks, including video depression detection [10,102], multimodal emotion recognition [47], natural language processing [103], image segmentation and classification [104], etc. In essence, the Transformer integrates the multihead attention mechanism and the residual module. Additionally, it incorporates position encoding to signify the relative or absolute positional relationships among elements within the sequence data. By using the self-attention mechanism to model global relationships, the Transformer architecture excels at learning long-range dependencies from input data and, more importantly, can be parallelized to make full use of GPU resources. Consequently, several transformer variants, such as tensorized encoder transformers [105], informer [106], and sparse binary Transformers [107], have been recently developed for processing time series sequence data.

Zhang et al. presented a deep learning approach known as the temporal difference-based graph transformer network (TDGNTN) to capture long-term temporal dependencies from input $PM_{2.5}$ data for air quality forecasting [100]. Fig. 5 illustrates the structure of the TDGNTN. Specifically, they focused on generating graph-structured data from input $PM_{2.5}$ data at various time points, even when the original data lacked an explicit graph structure. Next, they enhanced the self-attention mechanism by incorporating temporal difference characteristics and proposed

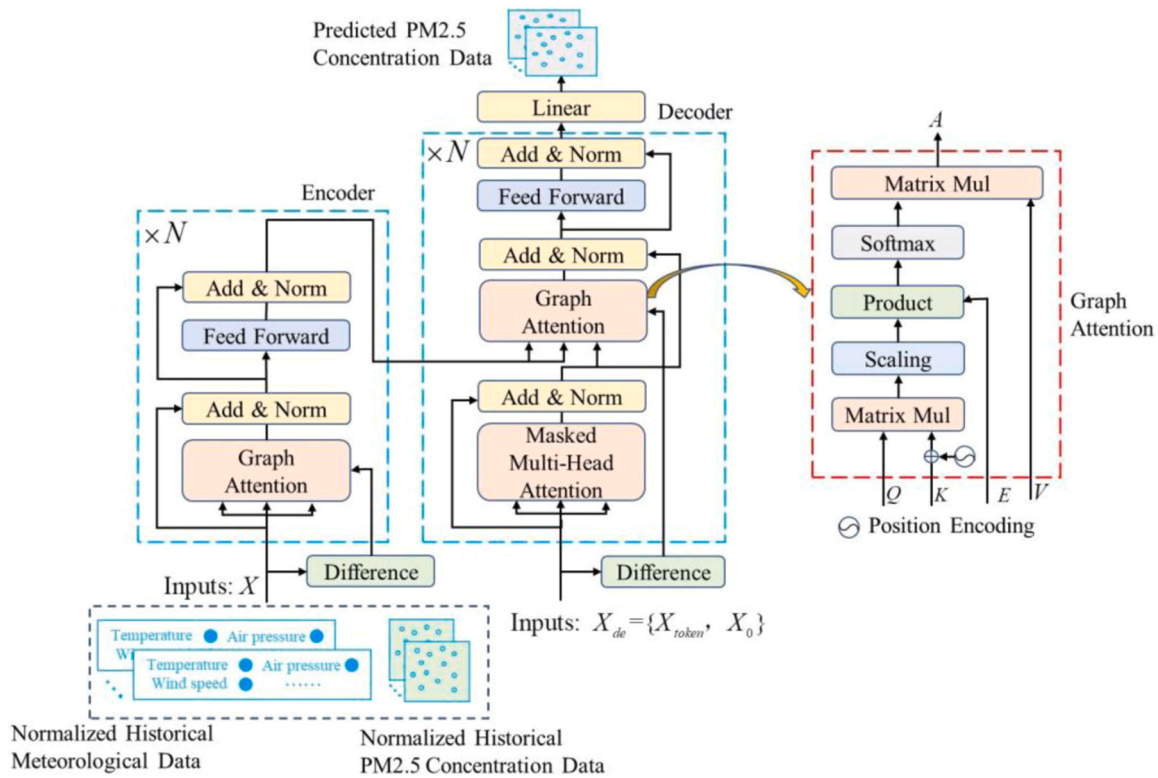


Fig. 5. The structure of the proposed TDGTM [100].

a novel graph attention mechanism, which was then integrated into the encoding and decoding layers of the TDGTM model. In this way, the TDGTM can capture long-term temporal dependencies and complex relationships from a graph perspective, improving the accuracy of predicting air quality at the PM_{2.5} level. Experimental results on two collected datasets in China indicate that the proposed method outperforms other methods for both short-term and long-term air quality forecasting. Zhang et al. proposed a lightweight deep learning approach based on sparse attention-based transformer networks (STNs) [7] to capture long-term dependencies and complex relationships from input data. The STN method consists of encoder and decoder layers and

employs multihead sparse attention mechanisms to decrease computational complexity. Feng et al. presented an encoder-decoder approach called the enhanced autoencoder (EnAutoformer) to enhance the air quality prediction performance [94]. Enhanced cross-correlation was employed to capture the temporal dependencies of the time series air data. Liang et al. provided a variant of the transformer called AirFormer for conducting nationwide air quality prediction tasks in China [108]. The proposed AirFormer consists of two stages: 1) a bottom-up deterministic stage for modeling spatiotemporal representations and 2) a top-down stochastic stage for learning the intrinsic uncertainty of air quality data. Iskandaryan et al. employed a GNN for predicting air

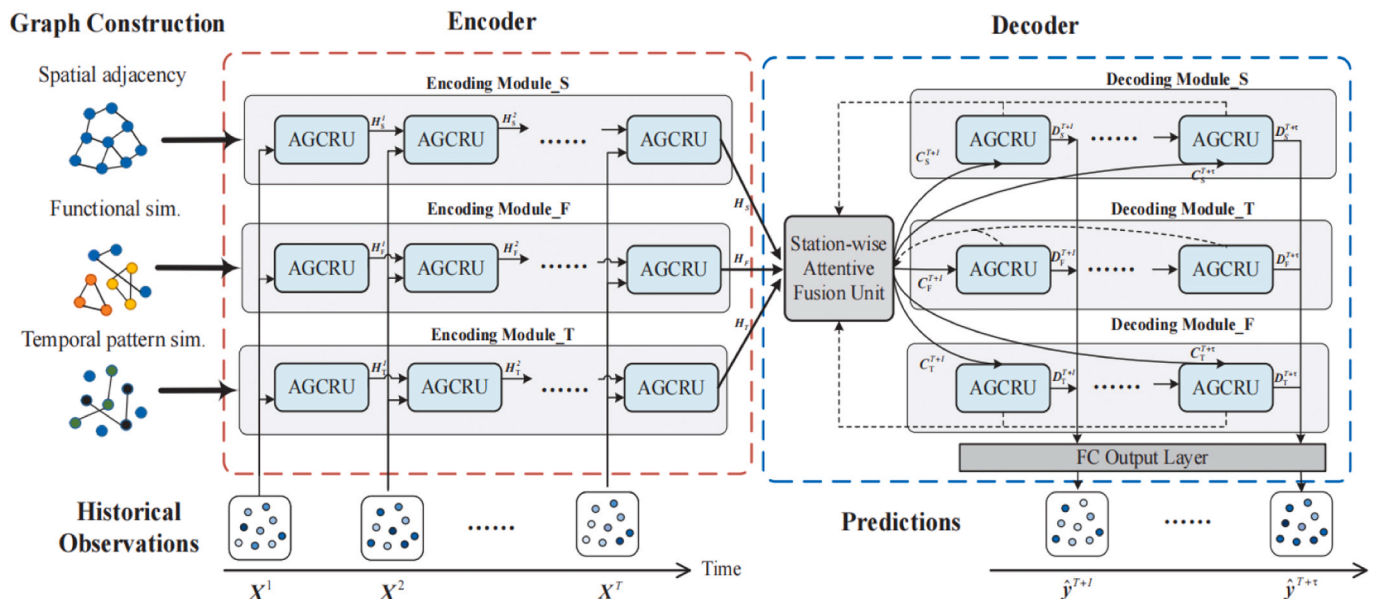


Fig. 6. The architecture of the proposed ATGCN[110].

quality in Madrid [109]. This model combines attention mechanisms, GRUs, and GCNs. The experimental results indicate that the proposed model outperforms other methods, such as the temporal graph convolutional network (TGCN), LSTM, and GRU. Wang et al. encoded the spatial adjacency, temporal pattern similarity, and functional similarity among air prediction stations in Beijing and Tianjin into a graph and proposed an attention temporal graph convolutional network (ATGCN) [110], as illustrated in Fig. 6. Within this network, graph convolution operations are integrated into a GRU through parallel encoding blocks. Furthermore, attention units are employed to ultimately generate multistep predictions for all sites within the decoding module. The results showed that ATGCN can effectively learn features across sites, resulting in enhanced prediction performance. Padhi et al. proposed hierarchical table bidirectional encoder representations from

transformers (BERT) based on transformers to learn tabular time series [111,112]. Air pollution concentrations are predicted through the learned encoding layer, and the effectiveness of the proposed method is demonstrated on real datasets. Wang et al. proposed a hybrid sequence-to-sequence (seq2seq) air quality prediction model with an attention mechanism to predict regional ground-level ozone concentrations [113]. This model is illustrated in Fig. 7. It utilizes the attention-based seq2seq model to extract temporal dependencies from input data as well as auxiliary data [114]. Then, it employed a deep neural network to extract spatial information. Ultimately, temporal and spatial models were combined to achieve the prediction results. The experimental results demonstrate that incorporating spatiotemporal correlation into the prediction model improved the prediction of regional ozone levels.

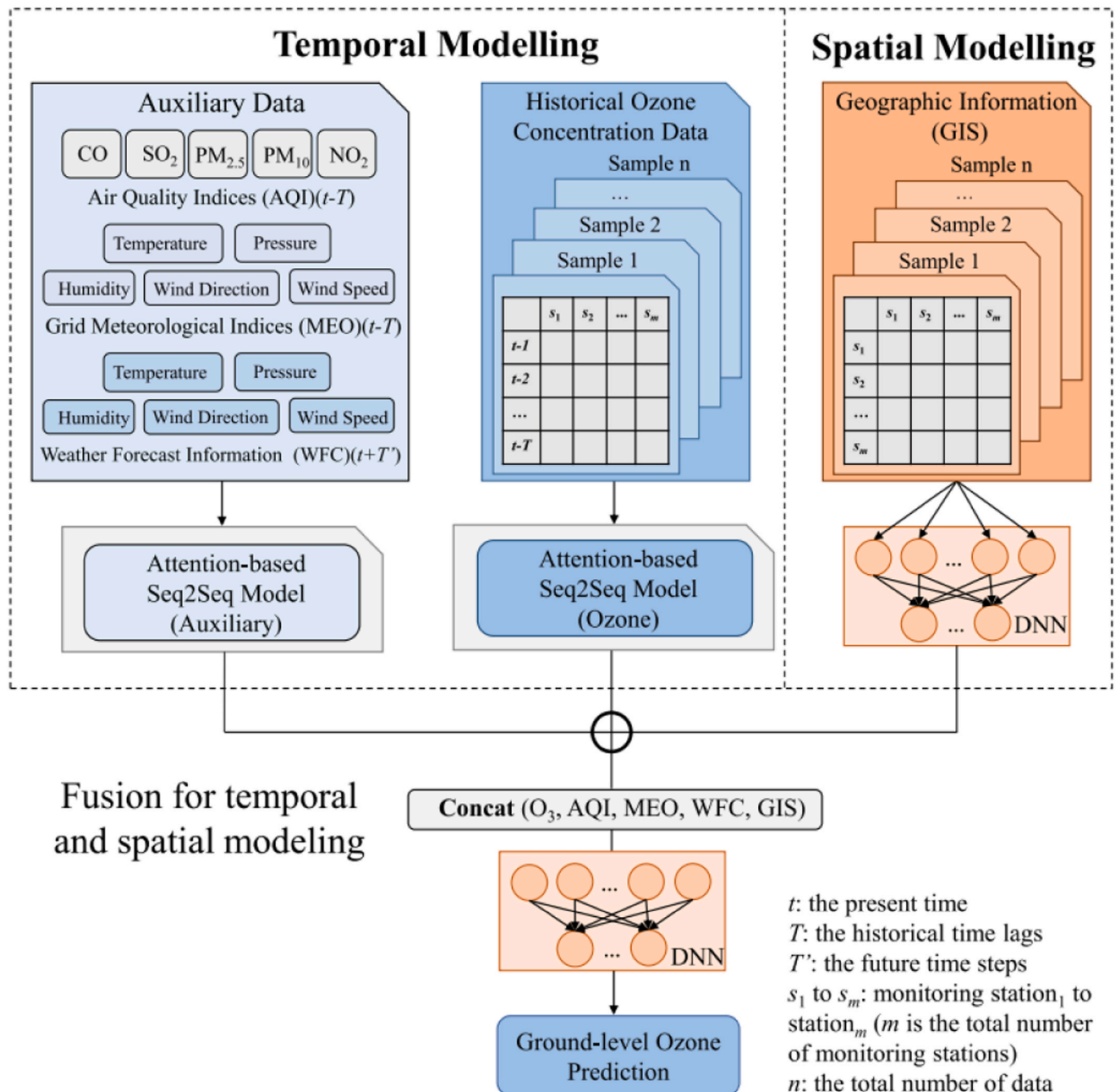


Fig. 7. The framework of the hybrid seq2seq model with attention[113].

5.6. Highlights in this section

Air pollution data are characterized by multiple features and variables and exhibit significant spatiotemporal characteristics. Attention mechanisms have demonstrated remarkable advantages when combined with multiscale and multitask deep learning strategies. This is attributed to their ability to capture long-term dependencies without being restricted by sequence length. Future progress in air quality modeling will involve the exploration of additional advanced deep learning models that integrate attention mechanisms for air quality prediction.

To summarize, Table 2 provides a comprehensive overview of air quality prediction methods based on deep learning, as mentioned above.

6. Challenges and opportunities

In recent years, the integration of deep learning techniques with air quality prediction has undergone substantial advancements. However, several challenges impede the continuous evolution of this field. Here, we summarize these challenges and present potential solutions and future directions for development, as described below.

- (1) Limitations in dataset coverage, data sparsity, noise, uncertainty, and bias.

The comprehensiveness and richness of relevant datasets have consistently remained crucial in the field of deep learning. Despite the gradual establishment of numerous air quality monitoring stations across various regions, comprehensive, large-scale datasets, including air pollution data, meteorological data, traffic conditions, altitude, and other variables, are still lacking. Moreover, air quality prediction research frequently necessitates the use of substantial historical datasets. However, due to the extended time span, the collected datasets often contain a substantial number of missing values, resulting in relative sparsity. In this case, data imputation is often needed during the pre-processing stage, inevitably impacting the subsequent prediction results. Therefore, it is necessary to explore a predictive model that can overcome or alleviate the problem of insufficient data.

To address the abovementioned issue, transfer learning approaches leveraging labeled data or knowledge structures from other correlated domains may enhance the learning effectiveness of target tasks [115,116]. Transfer learning techniques may address the challenges of high training costs and limited datasets in the context of air quality prediction. Ma et al. proposed a transfer learning-based stacked BiLSTM (TLS-BiLSTM) model to perform air quality prediction when processing new stations with few data [117]. This model involves pretraining and freezing the BiLSTM layer using the collected data from the existing stations, followed by fine-tuning the remaining network layers with the collected data from the new stations. Experiments conducted across all new air quality monitoring stations show that transfer learning strategies can significantly improve the prediction performance of new stations.

In addition, noise in air quality data may potentially affect air quality prediction. It is thus interesting to investigate the relationships between air and noise pollution in air quality prediction tasks [118]. In addition, uncertainty analysis is important for practical air quality management. To address this issue, developing uncertainty-aware deep learning models [119] for conducting air quality forecasting tasks may be feasible. Furthermore, bias correction [120] is also important for improving the air quality prediction performance. Therefore, exploring advanced bias correction methods for promoting air quality prediction performance is interesting.

- (2) High computational complexity, robustness, reliability, and interpretability of deep learning models.

Most existing deep learning models often possess a large

Table 2

Summary and comparison of air quality prediction methods based on deep learning.

Year	Literatures	Study Area	Methods	Performance
2017	Li et al.[89]	China	Geoi-DBN	MAE=8.54 RMSE=13.03 (h=24)
2018	Lin et al.[97]	Beijing, Los Angeles	GC-DCRNN	Beijing: MAE=36.62, RMSE=53.31 (h=24) Log Angeles: MAE=16.03, RMSE=21.08 (h=24)
2019	Tao et al.[75]	Beijing	CBGRU	MAE=10.47, RMSE=14.53 (h=1)
2019	Wu et al.[86]	North China	MSSTN	MAE=17.50, RMSE=29.87 (h=24)
2020	Wang et al.[98]	China	PM _{2.5} -GNN	MAE=15.91, RMSE=20.16 (h=72)
2020	Wang et al.[113]	Beijing	Attention-based seq2seq	MAE=16.11, RMSE=22.08 (h=24)
2021	Zhang et al.[70]	Beijing	VMDBiLSTM	MAE=7.12, RMSE=9.29 (h=1)
2021	Jin et al.[72]	Beijing	MTMC-NLSTM	MAE=0.88, RMSE=1.17 (h=1)
2021	Retta et al.[76]	Beijing	TCN	MAE=0.91, RMSE=2.18 (h=1)
2021	Han et al.[80]	Beijing, Shanghai	MasterGNN	Beijing: MAE=27.45, MAPE=0.548 (h=1) Shanghai: MAE=16.51, MAPE=26.5 (h=1)
2021	Huang et al.[87]	Beijing	SpAttRNN	MAE=57.36, RMSE=82.41 MAPE=0.698 (h=24)
2021	Yeo et al.[88]	Seoul	CNN-GRU	RMSE=10.08 (h=one year)
2021	Ge et al.[95]	Beijing	MST-GCN	MAE=18.44, RMSE=26.49 (h=24)
2021	Wang et al.[110]	Beijing, Tianjin	ATGCN	Beijing: MAE=11.25, RMSE=18.61 (h=1) Tianjin: MAE=14.27, RMSE=20.90 (h=1)
2021	Padhi et al.[111]	Beijing	BERT	RMSE=32.80 (h=1)
2022	Pruthi et al.[78]	Delhi	Nature-inspired deep learning	RMSE=10.80 (h=24)
2022	Xiao et al.[90]	China	DP-DDGCN	MAE=12.09, RMSE=15.19 (h=18)
2022	Gilik et al.[91]	Barcelona	CNN+LSTM	RMSE=0.07 (h=1)
2022	Faraji et al.[92]	Tehran	3DCNN-GRU	RMSE=15.21 (h=24)
2022	Huang et al.[93]	Beijing	EMD-IPSO-LSTM	MAE=4.02, RMSE=7.11 MAPE=8.07 (h=1)

(continued on next page)

Table 2 (continued)

Year	Literatures	Study Area	Methods	Performance
2023	Fang et al. [79]	Beijing	IVLSTM-MCMR	MAE=31.31, RMSE=33.04 (h=24)
2023	Zhao et al. [94]	Shanghai	SAC-QBSO-BiLSTM	MAE=10.16, RMSE=13.23 MAPE=25.53 (h=1)
2023	Iskandaryan et al. [109]	Madrid	A3T-GCN	MAE=15.91, RMSE=19.85 (h=24)
2023	Zhang et al. [100]	Beijing	TDGTN	MAE=11.06, RMSE=18.51 MAPE=22.91 (h=1)
2023	Zhang et al. [7]	Beijing	Sparse attention-based transformer networks	MAE=11.13, RMSE=19.04 (h=1)
2023	Elbaz et al. [99]	Saudi Arabia	ResNet-ConvLSTM	MAPE=13.57 (h=1)
2023	Feng et al. [94]	Yuzhong, China	EnAutoformer	MAE=0.538, RMSE=0.739 (h=24)
2023	Luo et al. [74]	Baoding, China	ARIMA-WOA-LSTM	MAE=5.572, RMSE=10.34 (h=1)
2023	Liang et al. [108]	China	Airformer	MAE=16.03, RMSE=32.36 (h=24)

number of network parameters, resulting in high computational complexity. However, due to the time-sensitive nature of air quality prediction, quick and accurate predictions are needed. Therefore, addressing the effective reduction in computational complexity in deep learning models is a crucial issue for future research. Additionally, an alternative strategy is to utilize dynamic neural networks [121,122], which can dynamically and adaptively adjust their structures and parameters based on input data. This approach can effectively reduce computational costs and enhance the versatility of dynamic neural networks. Air pollution data, which are typical time series sequence data, can benefit from dynamic data integration [123], which is an important direction for real-time forecasting.

Furthermore, the robustness, reliability, and interpretability of deep learning models are essential considerations in the field of deep learning. Robustness refers to a deep learning model's ability to make reliable decisions even in the presence of noise. Adversarial attack models [124] offer potential solutions to enhance the robustness of deep learning models. Reliability involves ensuring that a network converges successfully and stably during the training process [125]. Typical strategies such as regularization and normalization can be employed to ensure the reliability of deep learning models. Interpretability, on the other hand, aims to explain the working mechanisms of deep learning models. Both attributed-based and non-attributed-based methods [126] are popular approaches for providing explanations within deep learning models.

- (3) Exploring the deep relationships among air quality monitoring stations.

Existing deep learning models for air quality prediction typically focus on optimizing the objective function of a specific prediction task, often overlooking potential nonlinear spatial correlations among air quality monitoring stations. Effectively harnessing the shared characteristics of closely situated monitoring stations in adjacent regions is crucial in addressing this issue. One potential solution lies in adopting multitask learning (MTL) [127], which involves simultaneously learning multiple interrelated tasks. MTL offers the advantage of mitigating the

problem of local minima that single-task learning may encounter through task interactions. Additionally, the incorporation of multiple correlated auxiliary tasks has the potential to enhance the prediction performance for primary tasks [128]. In the context of air pollution prediction, data are often collected from densely distributed monitoring stations that exhibit numerous interdependencies. By applying MTL, it becomes possible to capture the shared characteristics among different tasks and integrate information from various sources effectively. This approach facilitates more accurate prediction of the complex interactions among air pollutants, ultimately leading to improved prediction performance. Consequently, MTL can serve as a valuable tool for exploring the deep relationships among different air quality monitoring stations.

- (4) Considering more factors for air quality prediction.

Various factors exert a significant influence on air pollution data, including the effects of high wind speeds, which can reduce PM_{2.5} concentrations, the impact of high humidity, which tends to increase air pollution levels, and the correlation between high atmospheric pressure and improved air quality, etc. These weather-related features play a crucial role in air quality prediction. Additionally, the emergence of the COVID-19 pandemic has led to noteworthy changes in human activity patterns that affect air pollution [94,129]. Furthermore, geographic terrain information and traffic data have been identified as variables that influence air quality prediction results [130]. In the future, exploring the integration of various factors, including weather data, the impact of the COVID-19 pandemic, satellite imagery, remote sensing data, terrain data, and points of interest can provide valuable insights into the collective impact of these factors on air quality prediction. It is important to note that data dynamics and factors such as government policies and medical treatment efficiency may necessitate consideration when modeling air quality prediction, warranting further investigation in the future. Additionally, it would be interesting to assess the performance and impact of these deep learning models on air quality management and policymaking in the future.

7. Conclusion

Air quality prediction is a multidisciplinary field that has attracted significant attention, especially with the emergence of deep learning. Deep learning has become a popular choice for air quality prediction due to its ability to uncover complex nonlinear relationships within extensive datasets. This review commences with an introduction to air pollution datasets, followed by an in-depth exploration of conventional air quality prediction methods and the utilization of deep learning techniques in this context. Additionally, this study provides a comprehensive overview of recent advancements in the application of deep learning methods for air quality prediction. Finally, the review outlines prospective directions for the field's development and offers an analysis of the challenges and opportunities in air quality prediction tasks.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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